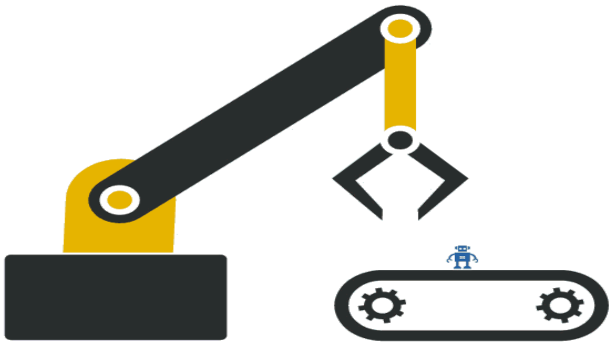
The Hack Clowns



EfenSys, an efficient Robotic Arm Solution for Manufacturing Industry

Team Members:

Israfil Shaheen Arannya
Cantonment Public School and College, Rangpur

Atiqur Rahman
Dania College, Dhaka

Mir Muhammad Abidul Haq Ahnaf
Monipur High School and College, Dhaka

Index

Abstract:

In the manufacturing industry, Robotics and the Internet of Things (IoT) solutions have become increasingly popular due to their numerous benefits, such as enhanced production efficiency, product quality, and workplace safety. A critical role is played by Robotic Arms in this sector. However, a major challenge faced by traditional robotic arms is the **need for hard coding for each new task**, which increases the cost of production and also it is time-consuming. So **The Hack Clowns** developed a robotic arm using the **JRC Board** in order to overcome these challenges. We developed a complete system that can reduce hard coding and time whenever a robotic arm needs to perform something different than its usual moves. This system can be easily programmed by authorized persons using an application, reducing the need for hard coding and increasing production efficiency. This can be helpful for third-world countries like ours.

Problems Statement:

1. Due to the repetitive hardcoding of robots for each new task, production costs increased.
2. Inefficient use of robots as they are limited to a specific task and cannot be easily adapted for other purposes.

3. Difficulty in programming and operating robots, requiring a specialized person with skill and knowledge.

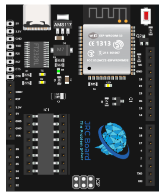
4. Lack of flexibility in the manufacturing process, leading to increased downtime and decreased productivity.

Our Solution:

The Hack Clowns team understands the challenges faced by the manufacturing industry, which is why we developed the **"EfenSys"** system using **JRC Board**, a cutting-edge solution that addresses these issues. EfenSys utilizes a user-friendly GUI application to instruct a versatile and adaptable robot over the internet, reducing the need for repetitive hardcoding and increasing production efficiency. A graphical interface makes programming easy and intuitive, eliminating the need for specialized personnel. With EfenSys, manufacturers can enjoy enhanced flexibility and decreased downtime, leading to increased productivity and reduced costs.

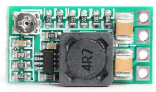
Hardware:

1. **JRC Board:** JRC board is a microcontroller board based on the ESP-WROOM-32. It has both Wi-Fi & Bluetooth facility builtin. This is the brain of our project.



2. **Servo Motor:** Servo motor will be used for the movement of our robotic Arm. We used 6 servos for this project.

3. **Buck Module:** We will use this module to step down the voltage from 12V to 5V. This 5V will power the JRC Board.



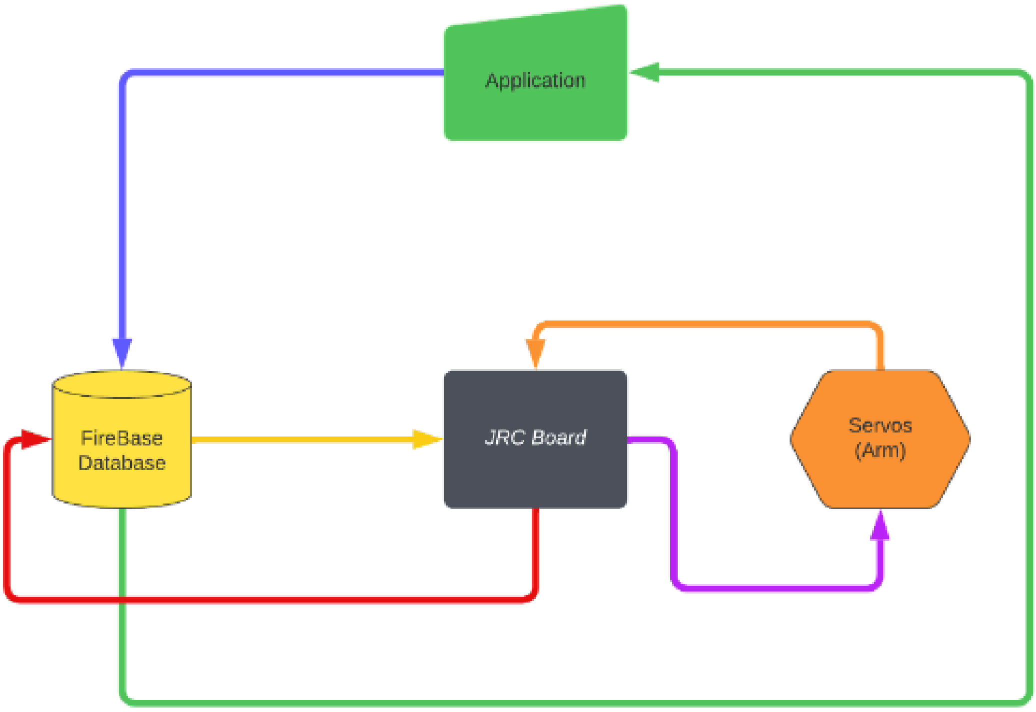
4. **12V SMPS:** It's a 12V 5A power supply. This will power up the whole EfenSys system.

Software:

1. **Arduino IDE:** We will use this Software to program our JRC Board.

2. **Google Firebase:** This will be the backend Server of our project.

Functional Diagram:



Functionality:

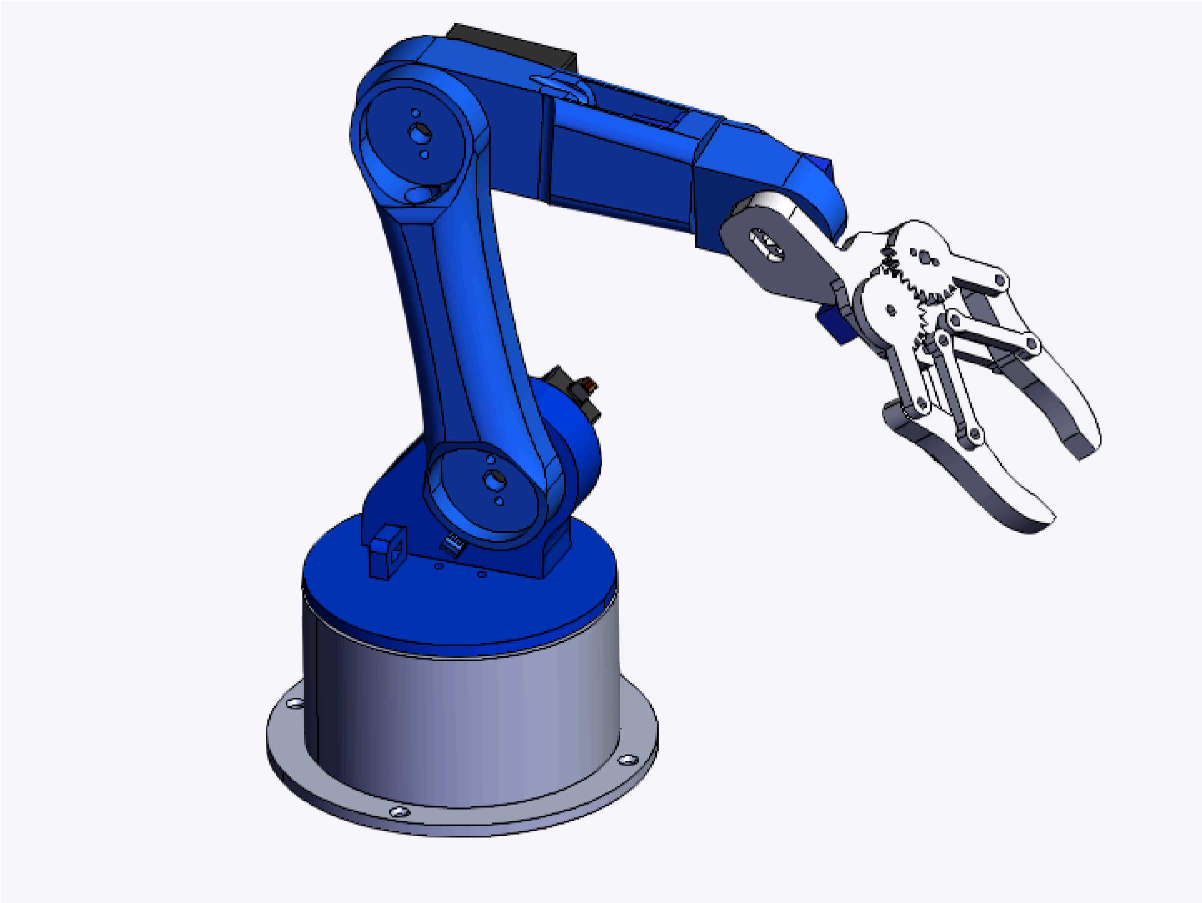
1. In our application there will be a graphical interface by which one can set the movement of the robot.

2. After setting, the instructions will be published to the database and the JRC Board will receive the data.

3. After receiving the data from firebase, the JRC Board will process the data and initiate the movement of the robot according to the received data.

4. If one wants to use this robot for another task he can easily re-instruct the robot in the same way.

3D Model:



[Click Here](#) for the 3D Model View.*[Credit: How to Mechatronics]*

N.B: We can change this design in the future for improvements and mechanical aspects to make it more functional.

Technical Specifications:

CPU	32 Bit
RAM	520 KB
Input Voltage	12V
Degrees Of Freedom	5 DOF
Max Height Reach	25cm
Max Weight Capacity	1.6 Kg

Future Improvements:

The future of manufacturing is here with EfenSys! The robotic arm is just a demo of the game-changing system. With this system, we will revolutionize the way industrial robots are used, making them more efficient and versatile than ever before. And that's not all! We'll introduce cutting-edge "**Digital Twin Technology**" that will allow manufacturers to simulate their industrial robots virtually. That means they can optimize their processes with ease, ensuring maximum efficiency and productivity before even implementing them.

Cost Estimation:

Product Name	Price (Tk)
MG996R Servo Motor * 3	1000/-
SG90 Servo Motor * 2	240/-
5V 5A SMPS * 1	450/-
Total Cost	**1690/-**

It will cost around 400,000/- to 500,000/- BDT to implement this technology in real life including the Arm itself.

Dependencies:

1. Uninterrupted Internet connection.
2. Good power supply.
3. Implementation in industry.